

CS4100
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Cracking the Code: AI for NYTimes Pips

A Comparative Study of CSP and Local Search Algorithms

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The background features a light beige color with stylized, overlapping illustrations of newspapers and documents. The newspapers are depicted with wavy lines representing text columns and some solid black rectangular areas representing headlines or images. The documents are shown as simple, layered shapes in various shades of beige and brown, creating a sense of depth and a collage-like effect.

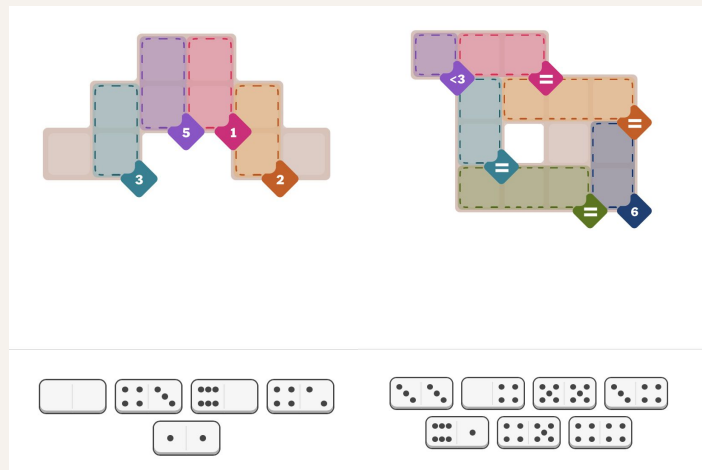
01

Introduction

What is NYT Pips? What is our project?

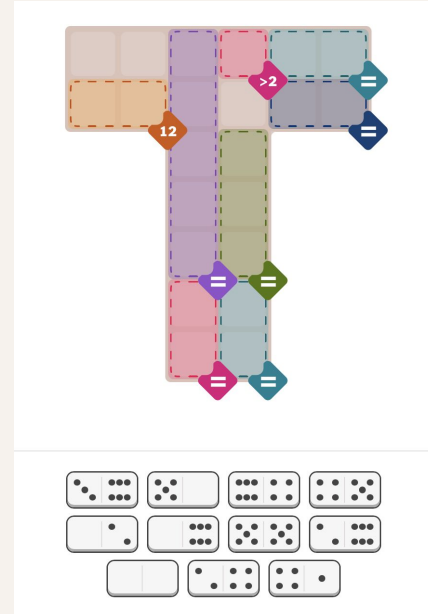
What is NYTimes Pips?

- New solo domino puzzle released by New York Times Games in August of 2025
- Players place dominoes on a grid under region-based constraints:
 - Sum
 - Equality
 - Inequality
- Difficulty scales from Easy (4–6 tiles) to Hard (14–16 tiles)



What is our Project?

- Build an optimal AI agent with the ability to solve all 3 difficulty levels of Pips
- Explore 2 AI strategies:
 - Local Search
 - Constraint Satisfaction Problem
- Evaluate which method best handles the constraint based grid game



The background features a light beige color with several stylized, overlapping illustrations of newspapers and papers. One newspaper is prominently shown in the top right corner, with its masthead and columns of text visible. Another newspaper is partially visible in the bottom left corner. A large, white, rounded rectangular shape serves as a backdrop for the main text, with some papers appearing to be tucked behind it.

02

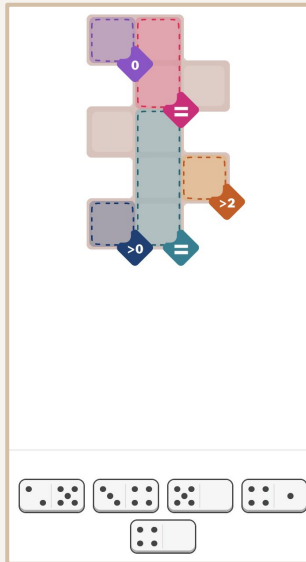
Strategy

How did we build this?

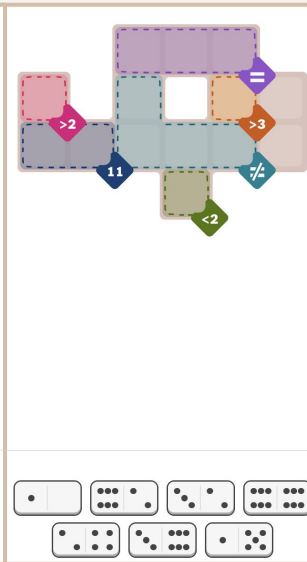
Building the Project

- Created mock versions of Pips for testing & solving
- Converted 200+ past Pips games into JSON objects
- Solved with two AI agents
- Created GUI for visualization

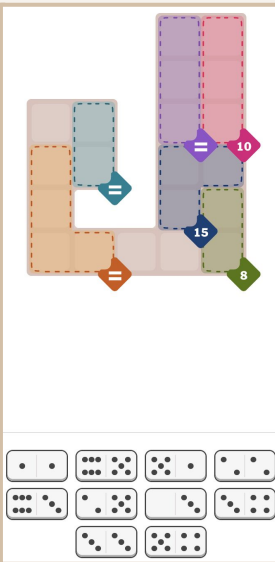
Easy



Medium



Hard



JSON Objects Explained

Board Constraints

Domino Format

Available Dominos

```
{
  "id": 126,
  "constructors": "Heidi Erwin",
  "dominos": [
    [
      2,
      2
    ],
    [
      2,
      3
    ],
    [
      5,
      5
    ],
    [
      0,
      2
    ]
  ],
  "regions": [
    {
      "indices": [
        [
          0,
          3
        ],
        [
          1,
          3
        ]
      ],
      "type": "equals"
    },
    {
      "indices": [
        [
          1,
          0
        ],
        [
          1,
          1
        ],
        [
          2,
          1
        ],
        [
          2,
          2
        ]
      ],
      "type": "sum",
      "target": 6
    },
    {
      "indices": [
        [
          2,
          3
        ]
      ],
      "type": "empty"
    },
    {
      "indices": [
        [
          3,
          1
        ]
      ],
      "type": "less",
      "target": 3
    }
  ]
}
```

Constraint Indices

Approach & Algorithms Used

Local Search

- Random placement
- Simulated annealing
- Moves: swap, flip, re-tile
- Temperature-based probability
- Cost function for constraint violations

CSP

- Dominoes placed one at a time with backtracking
- Forward checking for remaining constraints
- Pruning strategies

The background features a stylized illustration of a newspaper. A large, white, rounded rectangular page is centered, partially obscuring the newspaper underneath. The newspaper has a tan color and contains several columns of text represented by wavy lines. There are some black rectangular shapes at the top of the newspaper, possibly representing images or headlines. The overall style is clean and modern.

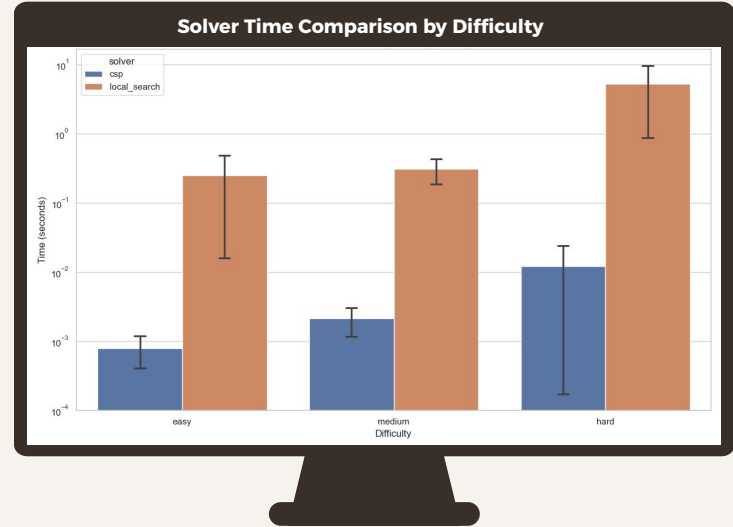
03

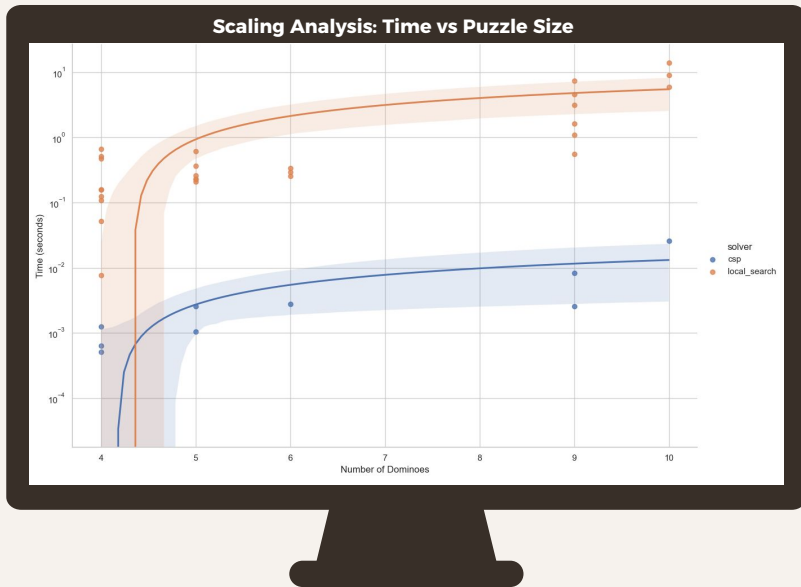
Results

Empirical Performance Analysis: CSP vs Local Search

CSP Dominates at Every Difficulty Level

- Easy: 1ms vs 250ms (250x faster)
- Medium: 1ms vs 250ms (250x faster)
- Hard: 20ms vs 5-10s (400x faster)



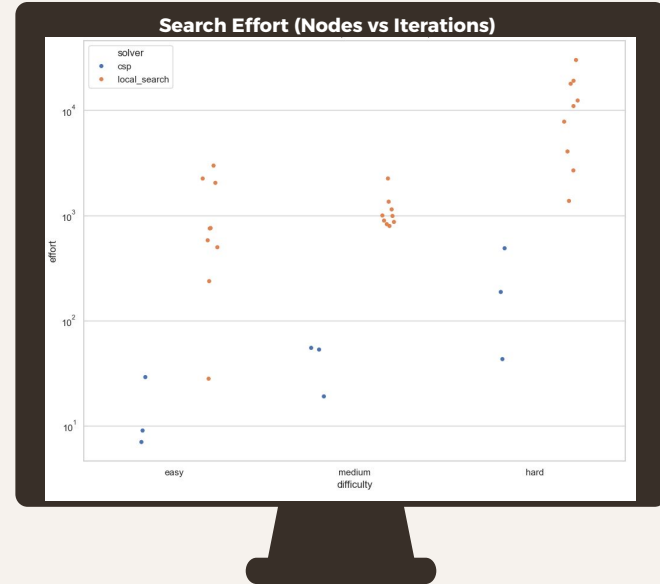


Scaling Reveals Exponential Crisis

- CSP: Linear growth (2x puzzle = 2x time)
- Local Search: Exponential explosion
- 15 dominoes: <1 second vs >10 minutes

Search Efficiency: 100x Difference

- CSP: 10–500 nodes with backtracking
- Local Search: 1,000–30,000 iterations
- Surgical precision vs trial-and-error

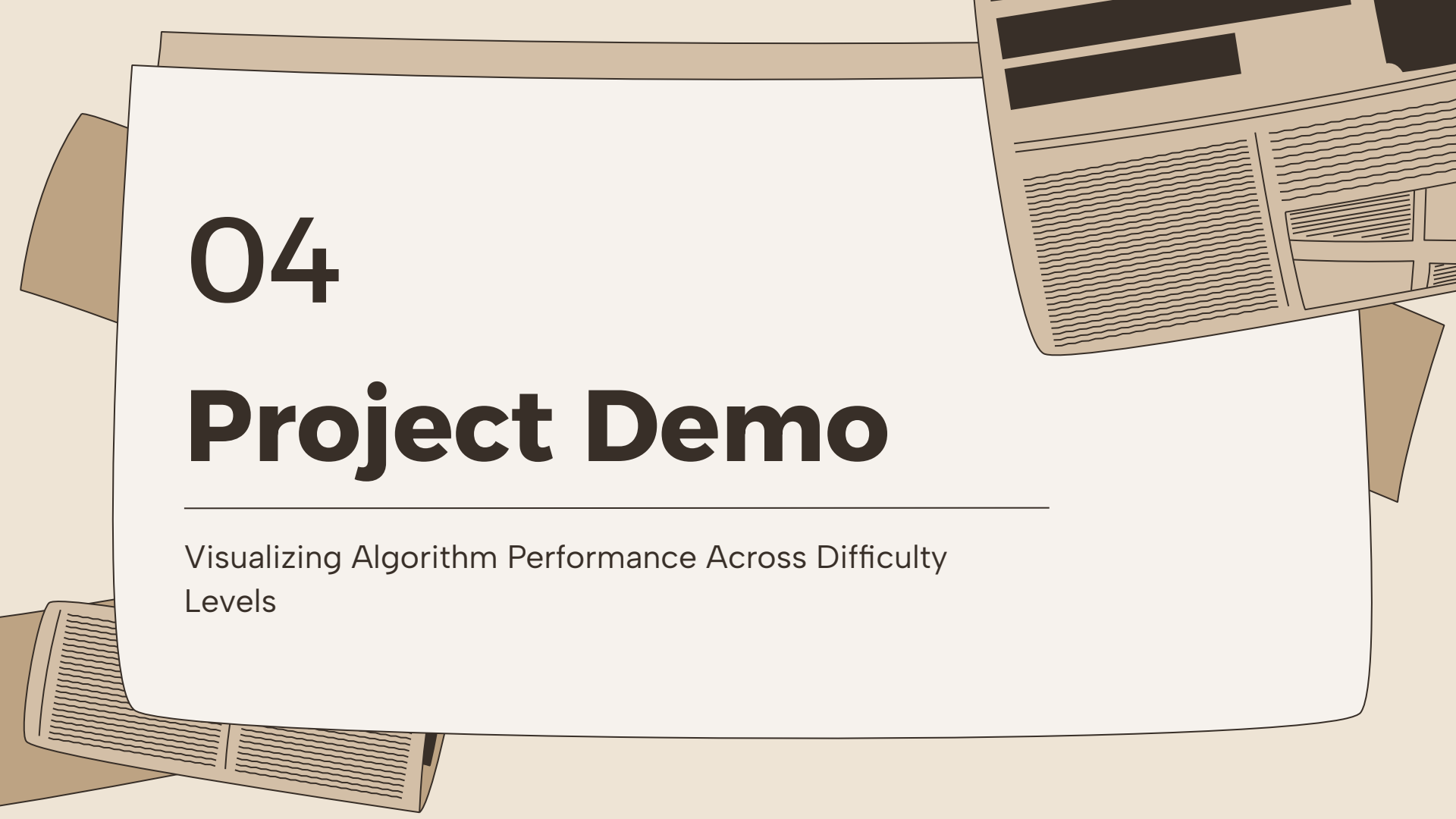




Whoa! Hypothesis overturned!

Expected: Local Search excels on constrained problems

Reality: Interconnected constraints favor systematic search

The background features a light beige color with several overlapping paper elements. A large white rectangular card is the central focus. To its right, a portion of a newspaper is visible, showing columns of text and a small image. Below the white card, another piece of paper with wavy lines is partially visible. The overall style is clean and modern.

04

Project Demo

Visualizing Algorithm Performance Across Difficulty Levels

Live Demonstration

<https://youtu.be/ME8ZCReth0w>

Easy Puzzle Performance

NYTimes Pips AI Solver

Controls

Load Puzzle:

Difficulty:

Date:

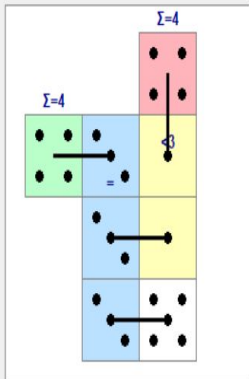
Solver Settings:

Algorithm:

Timeout (s):

☐ Solved!

Puzzle Board



Easy - 2025-11-19 | 4x3 | 4 dominoes | 4 constraints

Available Dominoes



Statistics & Logs

=== Puzzle Info ===

Board: 4x3
Total cells: 8
Dominoes: 4
Regions: 4
Filled: 4/4

Complete: Yes
Valid: Yes

```
[00:16:29] Loading easy puzzle for
2025-11-19...
[00:16:29] Puzzle loaded successfully!
[00:16:29] - Board size: 4x3
[00:16:29] - Dominoes: 4
[00:16:29] - Regions: 4
[00:16:36]
```

```
=====
[00:16:36] Starting CSP solver...
[00:16:36] Timeout: 30.0s
[00:16:36] ✓ Solution found in 0.000s!
```

```
Statistics:
[00:16:36] nodes_explored: 14
[00:16:36] backtracks: 9
[00:16:36] forced_moves: 0
```

NYTimes Pips AI Solver

Controls

Load Puzzle:

Difficulty:

Date:

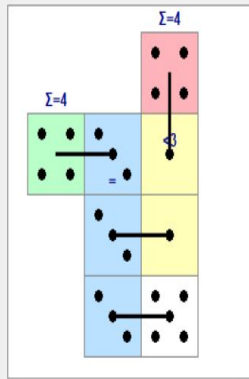
Solver Settings:

Algorithm:

Timeout (s):

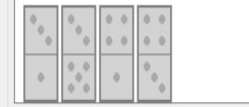
☐ Solved!

Puzzle Board



Easy - 2025-11-19 | 4x3 | 4 dominoes | 4 constraints

Available Dominoes



Statistics & Logs

=== Puzzle Info ===

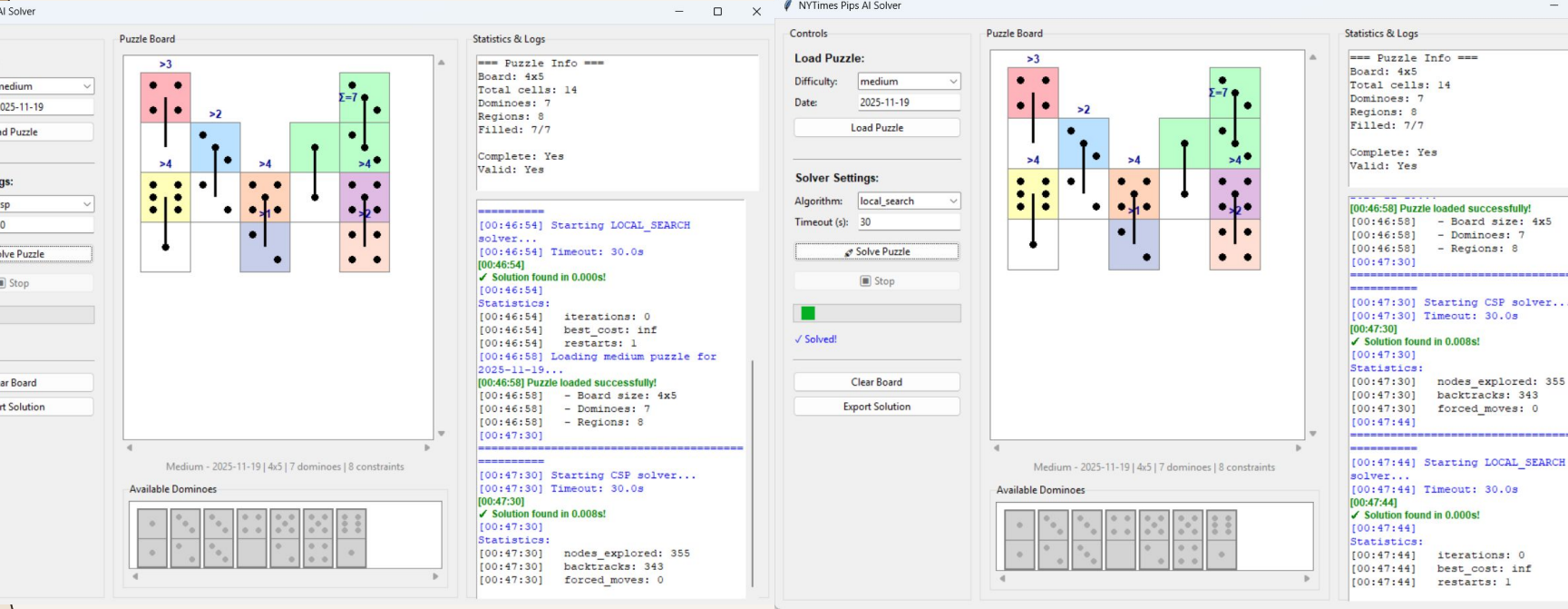
Board: 4x3
Total cells: 8
Dominoes: 4
Regions: 4
Filled: 4/4

Complete: Yes
Valid: Yes

```
=====
[00:16:36] Starting CSP solver...
[00:16:36] Timeout: 30.0s
[00:16:36] ✓ Solution found in 0.000s!
```

```
Statistics:
[00:16:36] nodes_explored: 14
[00:16:36] backtracks: 9
[00:16:36] forced_moves: 0
[00:16:54] Loading easy puzzle for
2025-11-19...
[00:16:54] Puzzle loaded successfully!
[00:16:54] - Board size: 4x3
[00:16:54] - Dominoes: 4
[00:16:54] - Regions: 4
[00:16:58]
```

Medium Puzzle Performance



Hard Puzzle: The Decisive Gap

NYTimes Pips AI Solver

Controls

Load Puzzle:

Difficulty:

Date: 2025-11-19

Solver Settings:

Algorithm:

Timeout (s): 30

☒ Solved

Puzzle Board

Statistics & Logs

==== Puzzle Info ====

Board: 5x8

Total cells: 26

Dominos: 13

Regions: 9

Filled: 13/13

Complete: Yes

Valid: Yes

```
=====
[00:47:44] Starting LOCAL_SEARCH
solver...
[00:47:44] Timeout: 30.0s
[00:47:44]
[00:47:44] ✓ Solution found in 0.000s!
[00:47:44]
Statistics:
[00:47:44] iterations: 0
[00:47:44] best_cost: inf
[00:47:44] restarts: 1
[00:49:03] Loading hard puzzle for
2025-11-19...
[00:49:03] Puzzle loaded successfully!
[00:49:03] - Board size: 5x8
[00:49:03] - Dominos: 13
[00:49:03] - Regions: 9
[00:49:26]
=====
[00:49:26] Starting CSP solver...
[00:49:26] Timeout: 30.0s
[00:49:26]
[00:49:26] ✓ Solution found in 0.061s!
[00:49:26]
Statistics:
[00:49:26] nodes_explored: 4721
[00:49:26] backtracks: 4707
[00:49:26] forced_moves: 0
```

Hard - 2025-11-19 | 5x8 | 13 dominos | 9 constraints

Available Dominos

NYTimes Pips AI Solver

Controls

Load Puzzle:

Difficulty:

Date: 2025-11-19

Solver Settings:

Algorithm:

Timeout (s): 300

Failed to solve

Puzzle Board

Statistics & Logs

==== Puzzle Info ====

Board: 5x8

Total cells: 26

Dominos: 13

Regions: 9

Filled: 0/13

Complete: No

Valid: No

```
=====
[00:49:26] Starting CSP solver...
[00:49:26] Timeout: 30.0s
[00:49:26]
[00:49:26] ✓ Solution found in 0.061s!
[00:49:26]
Statistics:
[00:49:26] nodes_explored: 4721
[00:49:26] backtracks: 4707
[00:49:26] forced_moves: 0
[00:49:52] Loading hard puzzle for
2025-11-19...
[00:49:52] Puzzle loaded successfully!
[00:49:52] - Board size: 5x8
[00:49:52] - Dominos: 13
[00:49:52] - Regions: 9
[00:49:55]
=====
[00:49:55] Starting LOCAL_SEARCH
solver...
[00:49:55] Timeout: 300.0s
[00:54:55]
[00:54:55] ✗ No solution found within timeout
[00:54:55]
Statistics:
[00:54:55] iterations: 2548563
[00:54:55] best_cost: 17.0
[00:54:55] restarts: 1849
```

Hard - 2025-11-19 | 5x8 | 13 dominos | 9 constraints

Available Dominos

The background features a light beige color with several stylized, overlapping illustrations of newspapers and papers. One newspaper is prominently shown in the top right corner, with its masthead and columns of text visible. Another newspaper is partially visible in the bottom left corner. A large, white, rounded rectangular shape serves as a backdrop for the main text, with some brown paper-like shapes peeking out from behind it.

05

Conclusion

How can we move forward?

Key Takeaways

- CSP is the more optimal solver for NYT Pips
- Overlapping regional constraints align naturally with systemic search
- Local Search struggles because Pips penalizes partial progress
- Puzzle structure (not just difficulty) determines algorithm performance

[Back to puzzle](#) ✕



Congrats!

You finished an **easy** puzzle in **0:35**.

New puzzles released daily at midnight local time.

[Share Your Results](#)



[Play Today's Spelling Bee](#) >

What we learned about Pips?

- Pips behaves like a true constraint satisfaction problem, not a noisy optimization task
- Constraint propagation yields major gains (especially on Hard puzzles)
- Performance scales gracefully with puzzle size when constraints are leveraged
- “Guess and improve” strategies fail when all regions interact tightly

[Back to puzzle](#) ✕



Congrats!

You finished a **medium** puzzle in **1:44**.

New puzzles released daily at midnight local time.

[Share Your Results](#)



[Play Today's Spelling Bee](#) >

Broader Implications

- CSP-based solvers could generalize to scheduling, logistics and layout tasks
- Local Search still holds potential with hybrid or informed heuristics
- Pips is a useful benchmark puzzle for teaching multi-constraint AI

[Back to puzzle](#) ✕



Congrats!

You finished a **hard** puzzle in **2:24**.

New puzzles released daily at midnight local time.

[Share Your Results](#)



[Play Today's Spelling Bee](#) >

The background features a light beige color with several stylized, overlapping illustrations of newspapers and papers. One newspaper is prominently shown in the top right corner, with its masthead and columns of text visible. Another newspaper is partially visible in the bottom left corner. A large, white, rounded rectangular shape serves as a backdrop for the main text, with some paper scraps visible behind it.

06

Questions/Comments

Thank you for listening to our presentation